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Adaptive interference management in aerial/UAV-supported 5G networks using deep reinforcement learning

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ABSTRACT

In the current era of technological advancement, the use of drones has expanded in various fields. These fields include aerial photography, medical supply delivery to accident scenes, forestry, and agricultural monitoring. However, the presence of a large number of drones operating simultaneously within a shared airspace and common spectrum bands creates a significant challenge. This challenge lies in managing communication interference between these aircrafts. This research aims to address this problem by developing a framework based on deep reinforcement learning techniques to adaptively manage interference and make decisions according to the current state of the network. The Policy Proximal Optimization (PPO) algorithm is employed to respond to network conditions in real-time and dynamically select the most appropriate action. These actions include adjusting transmission power, repositioning, maintaining the current state, or reducing transmission power as needed. The proposed model achieved an average performance of 85.7%, representing a 23.4% improvement compared to the pre-optimization value. It also demonstrated a 13.8% improvement in high-interference scenarios and a 14.9% enhancement in signal-to-noise ratio. These results highlight the potential of deep reinforcement learning as an adaptive and effective solution for mitigating interference in drone networks.

1. Introduction

Network monitoring and traffic analysis have become indispensable in modern communication systems to ensure performance, reliability, and security [1–3]. The use of drones has become widespread across many sectors. They are used in various applications, including aerial photography, medical supply delivery, forest monitoring, and agricultural surveillance [4,5]. However, the presence of a large number of drones operating simultaneously within a shared airspace and common spectrum bands poses a significant challenge. Managing communication interference among these aircraft becomes increasingly complex under such conditions. As their transmitted signals interact with each other, substantial degradation in communication quality can occur [6]. This, in turn, leads to additional challenges, such as a low signal-to-interference-plus-noise ratio (SINR). This degradation adversely affects data transmission rates and communication reliability [7]. These challenges have highlighted the need to develop intelligent systems to mitigate interference.

Traditional interference management methods rely on static protocols developed through pre-programmed processes that define fixed parameters [8], such as static transmission power levels. These interactions are typically handled manually and in a simplified manner, based on predefined thresholds. This approach is impractical and unsuitable for dynamic drone environments, where the number of drones, their distribution, and operating frequencies continuously change.

In recent years, artificial intelligence technologies have emerged prominently and have been widely adopted across various fields. They have achieved promising results and are increasingly viewed as a potential solution to many challenges, including interference management [9]. Deep Reinforcement Learning (DRL) has emerged as a promising approach for addressing the problem of interference management [10]. It enables agents to learn optimal behaviors through direct interaction with their environment [11], gradually improving their decision-making policies over time using a reward-and-penalty mechanism. This model is well-suited to the interference management problem. The agent monitors network parameters such as SINR and interference levels. It then selects an appropriate action based on this data, such as increasing transmission power, repositioning, maintaining the current state, or decreasing power. The environment then moves into a new state and offers a reward that guides the learning process.

However, the successful application of DRL in this field requires careful design of its three core components: the observation space, the workspace, and the reward function [12,13]. A simple observation space that provides only an immediate SINR reading fails to capture the temporal trends required for proactive decision-making. In addition,

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a sparse reward function that penalizes only complete link failures does not provide sufficient guidance for the agent to learn precise and effective behaviors.

This paper addresses these complexities by proposing a framework for the adaptive mitigation of drone interference. The framework incorporates a comprehensive observation space that monitors current network metrics, historical trends, and operational context. This design enables the agent to better understand the network's trajectory and make more informed decisions. The proposed framework also includes an action space that allows transmission power adjustments and repositioning. It incorporates operational constraints that reflect real-world conditions. A multi-component reward function has also been developed to encourage appropriate behaviors. It incorporates dynamic weighting to balance competing objectives, such as throughput, energy efficiency, and interference reduction.

The rest of this paper is organized as follows. Section 2 presents the related work and reviews recent advances in UAV interference management and reinforcement learning-based optimization. Section 3 presents an overview of the proposed system and its problem formulation. Section 4 describes the deep reinforcement learning framework, including the PPO-based architecture that is used in this work. The scenario-adaptive reward engineering strategy is described in Section 5, including the design of the multi-objective reward function and adaptive rewarding mechanism. In Section 6, the dynamic action space and operational constraints governing UAV decisions are provided. Section 7 describes the intelligent exploration strategy employed to enhance policy diversity and stability. Section 8 presents the experimental results and provides a performance analysis, concluding with remarks. Finally, Section 9 gives the summary of the paper and future work.

1.1. Contributions and novelty

Traditional interference management methods rely primarily on fixed power control or predefined thresholds. Current DRL solutions, on the other hand, depend on fixed monitoring areas and fixed reward formulas. In contrast to these approaches, this paper presents a scenario-adaptive DRL framework for 5G drone-enabled networks. The main contributions of this work are summarized below:

1. The proposed framework introduces a scenario-aware reward engineering mechanism. This mechanism adjusts reward weights based on network conditions (high interference, low signal-to-noise ratio, and normal operation). This approach enables more accurate and realistic scenario-specific decision-making compared to static reward designs.
2. The proposed framework relies on an enhanced monitoring space that integrates real-time network metrics with temporal trend information and past actions. This mechanism allows the agent to monitor network evolution and changes over time, rather than relying on isolated snapshots of the network state.
3. The proposed framework is based on a dynamic exploration strategy to improve policy proximity (PPO). This strategy relies on entropy monitoring, which prevents premature convergence and improves policy robustness in dynamic drone environments.
4. This paper proposes a unified decision-making framework. The framework jointly improves transmission power control and drone repositioning under realistic operational constraints. This approach addresses the shortcomings of previous work that focused on only one control dimension.
5. The results were validated using a real-world, fifth-generation unmanned aerial vehicle (AERPAW) dataset, rather than relying on simulated data. This demonstrates the practical applicability and generalizability of the proposed framework beyond simulated environments.

Together, these contributions enhance the adaptability and robustness of DRL-based interference management. They also improve its practical relevance in 5G drone-supported networks.

2. Related work

Several studies have employed DRL frameworks to address various challenges. For example, study [14] utilized the MADDPG algorithm to dynamically optimize UAV positioning and Intelligent Reflecting Surface (IRS) configurations. Study [15] applied a Soft Actor-Critic (SAC) approach to minimize the Age of Information (AoI) for data collection from IoT devices. Studies such as [16,17] proposed iterative optimization approaches to address path planning problems. Specifically, they employed methods such as Block Successive Convex Approximation (BCSA) and Block Coordinate Descent (BCD).

Through a critical analysis of previous studies, several challenges and limitations have been identified that require further consideration. For example, studies such as [14,18] relied on idealized assumptions regarding channel state information, which are impractical in real-world applications. Studies [15,16] examined the computational complexity of their proposed solutions. This high complexity makes real-time implementation challenging, particularly in environments with shared and limited resources.

The proposed frameworks in studies [19,20] focused on adjusting UAV settings individually. However, they did not consider coordination among multiple agents operating within the same airspace. Other limitations include the assumption of constant UAV altitudes [21] and unoptimized power constraints [20]. Additionally, some approaches are sensitive to incomplete phase compensation within elements of the internal control system [19].

Although numerous studies have focused on improving drone communication over 5G networks using deep reinforcement learning, several limitations remain. Many of these approaches rely on idealized channel state information. Some employ fixed reward formulations that do not adapt to rapidly changing interference conditions. Others define limited action spaces that optimize either transmission power or location separately. In addition, several studies base their evaluation solely on artificial simulation environments. These limitations reduce their effectiveness in real-world applications. The proposed framework addresses these gaps by incorporating scenario-adaptive reward shaping, time-sensitive state representations, and dynamic exploration control. It also includes evaluation using real-world datasets and laboratory-based experiments. A comparative summary of previous methods is provided in Table 1.

3. System overview and problem formulation

In this paper, we rely on the development of a framework that models the UAV network as a multi-agent system. This avoids the limitations of previous studies that depended on individual UAV configurations. Those studies overlooked the coordination between multiple agents (UAVs) that can fly within the same airspace. In the proposed framework, each UAV would manage its transmission parameters based on local observations. It also draws from network information, other UAV signals, and historical context. These UAVs behave as individual entities that sense the local network environment. They also optimize their control policy. However, there would be no direct contact or sharing of parameters among these individual UAVs. The UAVs would collectively coordinate implicitly. There would be an environment through which this implicit coordination would occur. Each agent's actions affect aggregate interference levels, SINR, and network load observed by neighboring agents. The problem is formulated as a partially observable Markov decision process (POMDP) [22]. The main features of the proposed framework are:

- **State Space:** Network conditions (SINR, interference, load, UAV count).
- **Action Space:** Reduce power, maintain power, increase power, and reposition.

Table 1
Summary of related work and identified limitations.

Reference	Method	Main Objective	Limitations
[14]	MADDPG with UAV–IRS joint optimization	Optimize UAV positioning and IRS phase shifts	Assumes ideal channel state information; limited robustness to dynamic interference
[15]	Soft Actor-Critic (SAC)	Minimize Age of Information (AoI) for IoT data collection	High computational complexity; limited real-time applicability
[16]	BCSA-based iterative optimization	UAV path planning and coverage optimization	Computationally expensive; slow convergence in large-scale scenarios
[17]	Block Coordinate Descent (BCD)	Secrecy rate maximization	Requires centralized control; limited adaptability to dynamic environments
[18]	DRL-based throughput optimization	Maximize system throughput	Relies on idealized CSI assumptions; fixed reward formulation
[19]	DRL-based UAV control	BER minimization	Sensitive to phase compensation errors; ignores multi-UAV coordination
[20]	DRL for AoI optimization	AoI minimization	Unoptimized power constraints; single-agent focus
[21]	Multi-UAV optimization framework	Multi-UAV communication enhancement	Assumes constant UAV altitude; limited adaptability
Proposed Work	Scenario-adaptive PPO with dynamic exploration	Joint interference management, throughput, and SINR optimization	Addresses adaptive reward shaping, realistic CSI, multi-UAV coordination, and real-world evaluation

- **Reward Function:** A multi-objective reward function which developed to balance throughput and energy efficiency and mitigate interference.

System control operations are implemented dynamically based on several parameters, which are explained in formula (1):

$$s_{t+1} = f(s_t, a_t) + \omega_t \tag{1}$$

Where:
s_t: Network state.
a_t: Action.
ω_t: Environmental noise.

In the proposed framework, the UAV environment is assumed to be a Markov Decision Process (MDP) determined by a set of parameters, which are explained in formula (2).

$$M = (S, A, P, R, \gamma) \tag{2}$$

Where:
S: State space.
A: Set of UAV actions.
P: Probabilities of the state transition.
R: Reward function.
γ: Discount factor.

It is important to clarify the relationship between the POMDP formulation and the Markovian Decision Process (MDP)-based learning algorithm used in this work. The underlying drone network is inherently a partially observable Markovian Decision Process (POMDP) because each drone has only local information about the state of the global network. However, in DRL, the POMDP is approximated as an MDP. This is achieved by constructing an enhanced state representation that incorporates real-time observations, historical trends, and past actions. This is an approximation of the state of belief, enabling the agent to capture relevant temporal dependencies and mitigate partial observability. As a result, standard MDP-based policy optimization algorithms can be effectively applied while preserving the essential dynamics of the original POMDP.

At each timestep t the UAV observes a state $s_t \in S$, executes an action $\alpha_t \in A$ and receives a scalar reward $r_t = R(s_t, \alpha_t)$.

The action space is discrete:

$$A = \begin{cases} 0: & \text{Reduce Power} \\ 1: & \text{Maintain Power} \\ 2: & \text{Increase Power} \\ 3: & \text{Reposition UAV} \end{cases}$$

In this research, we used the dataset (*AERPAW 'Ericsson 5G NSA network RF & throughput measurements with UAV'*) [23]. This dataset includes radio frequency measurements taken from Ericsson's 5G network using a UAV in September 2023. The dataset includes a range of metrics such as RSRP, SINR, CQI, MCS, and throughput [23]. The experiment and measurements were conducted at the University of North Carolina at Centennial, which simulates an urban environment. They were also conducted at the Wheeler Lake site, which reflects a rural agricultural environment. Both sites are equipped to provide comprehensive 5G network coverage. During the flight, the UAV followed a predetermined, zigzag flight path [23].

4. Deep reinforcement learning framework

4.1. Proximal policy optimization (PPO) architecture

We used PPO [24], which ensure stability and convergence. The policy network defines the distribution of actions as follows in Eq. (3):

$$\pi_\theta(\alpha_t, s_t) = \text{Softmax}(f_\theta(s_t)) \tag{3}$$

The agent is trained using PPO, which is a clipped objective function shown in formula (4):

$$L^{CLIP}(\theta) = E_t\left[\min(r_t(\theta)\,\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t)\right] \tag{4}$$

Where:
$r_t(\theta) = \dfrac{\pi_\theta(\alpha_t, s_t)}{\pi_{\theta_{old}}(\alpha_t, s_t)}$
$\hat{A}_t$: Advantage estimate.
ϵ: The clipping range.
This function ensures stable gradient updates while maintaining policy improvement safeguards.
The PPO network is a multilayer perceptron network (MLP) [25] with four layers:

- Input_layer with 12 neurons.
- Hidden_layer1 with 64 neurons.
- Hidden_layer2 with 32 neurons.
- Output_layer with 4 neurons represents the action probability.

4.2. Enhanced observation space design

The observation vector has 12 dimensions of network showed in formula (5):

$$o_t = \left[\text{SINR}_t, I_t, L_t, N_{UAV,t}, P_t, \Delta\,\text{SINR}_t, \Delta I_t, \alpha_{t-1}, \alpha_{t-2}, C_P, C_R, \tau_R\right] \tag{5}$$

Where:
$SINR_t$: Signal-to-Interference-plus-Noise Ratio.
I_t: Interference level.
L_t: Network load.
$N_{UAV,t}$: UAVs count in the region.
P_t: Current power.
$\Delta SINR_t$ and ΔI_t: SINR and interference.
α_{t-1} and α_{t-2}: Previous actions.
C_P: Normalized power constraint.
C_R: Reposition readiness.
τ_R: Repositioning ratio.
Trend analysis is an enhanced weighted moving average with exponential smoothing, showed in formula (6):

$$Trend_t = \frac{x_t - \sum_{i=1}^{w} w_i x_{t-i}}{\max(|\sum_{i=1}^{w} w_i x_{t-i}|, 1)} \tag{6}$$

Where: The weights $w_i = 0.5 + \frac{(i-1)}{(w_{i-1}-1)}$ provide recency bias.

5. Scenario-adaptive reward engineering

5.1. Multi-objective reward function

As mentioned earlier, an adaptive reward function was developed, which integrates network performance metrics, operational constraints, and scenario-specific adjustments. The reward at timestep t is defined as shown in formula (7):

$$R_t = \phi_1 T_t, \phi_2 E_t, \phi_3 S_t, \phi_4 I_t, \phi_5 L_t, \phi_6 B_t, \phi_7 P_t, \phi_8 X_t \tag{7}$$

Where:
$T_t = (\log(1 + SINR_t))/10$: Normalized throughput.
$E_t = (SINR_t)/(I_t + \epsilon)$: Energy efficiency.
S_t: SINR-based reward.
I_t: Interference level penalty.
L_t: Network load penalty.
B_t: A status-dependent reward that takes a dynamic value based on the scenario.
P_t: Constraint penalty for violating power or reposition rules.
X_t: Exploration bonus for diverse behavior.
$\phi_1, \ldots, \phi_8$: adaptive weighting coefficients summarized in Table 2.
All reward components are normalized before aggregation to ensure that no single term dominates the learning process due to scale differences.

Table 2
Final reward weighting coefficients used in the proposed framework.

Coefficient	Value
ϕ_1	0.15
ϕ_2	0.15
ϕ_3	0.25
ϕ_4	−0.35
ϕ_5	−0.08
ϕ_6	0.8
ϕ_7	1.0
ϕ_8	1.0

5.2. Scenario classification and adaptive rewarding

We have identified three operational scenarios with varying reward values specific to each scenario according to its importance:

- **High interference scenario with $I > 25$ dB:**
  Optimal choice: Power reduction ($\alpha = 0$) or repositioning ($\alpha = 3$).
  Suboptimal choice: Power increase ($\alpha = 2$).
- **Low SINR Scenario with $SINR < 15$ dB:**
  Optimal choice: Power increase ($\alpha = 2$) or repositioning ($\alpha = 3$).
  Suboptimal choice: Power reduction ($\alpha = 0$).
- **Normal conditions:**
  Optimal choice: Maintain power ($\alpha = 1$).
  Suboptimal choice: Any unnecessary action.

The bonus B_t is dynamically adjusted based on the identified scenario condition as follows:

$$B_t = \begin{cases} +0.6 & \text{Any optimal action in critical scenario} \\ +0.4 & \text{If normal conditions scenario \& action is maintained} \\ -0.8 & \text{Any suboptimal action in any scenario} \end{cases}$$

6. Dynamic action space and constraints

6.1. Action space definition

As we mentioned in Section 3:

$$A = \begin{cases} 0: & \text{Reduce Power} \\ 1: & \text{Maintain Power} \\ 2: & \text{Increase Power} \\ 3: & \text{Reposition UAV} \end{cases}$$

Below we define each action:
Action 0: Reduce transmission power as: $P \leftarrow P \times 0.8$
Action 1: Maintain power.
Action 2: Increase transmission power as: $P \leftarrow P \times 1.25$
Action 3: Reposition UAV with cooldown time = 5 steps.

6.2. System dynamics model

The effects of power modulation on the SINR are as follows:

$$SINR_{\text{new}} = SINR_{\text{current}} \times \begin{cases} 0.85 \text{ if action 0 (Reduce power)} \\ 1.15 \text{ if action 2 (Increase power)} \\ 1.25 \text{ if action 3 (Reposition } UAV\text{)} \\ 1 \text{ if action 1 (Maintain power)} \end{cases}$$

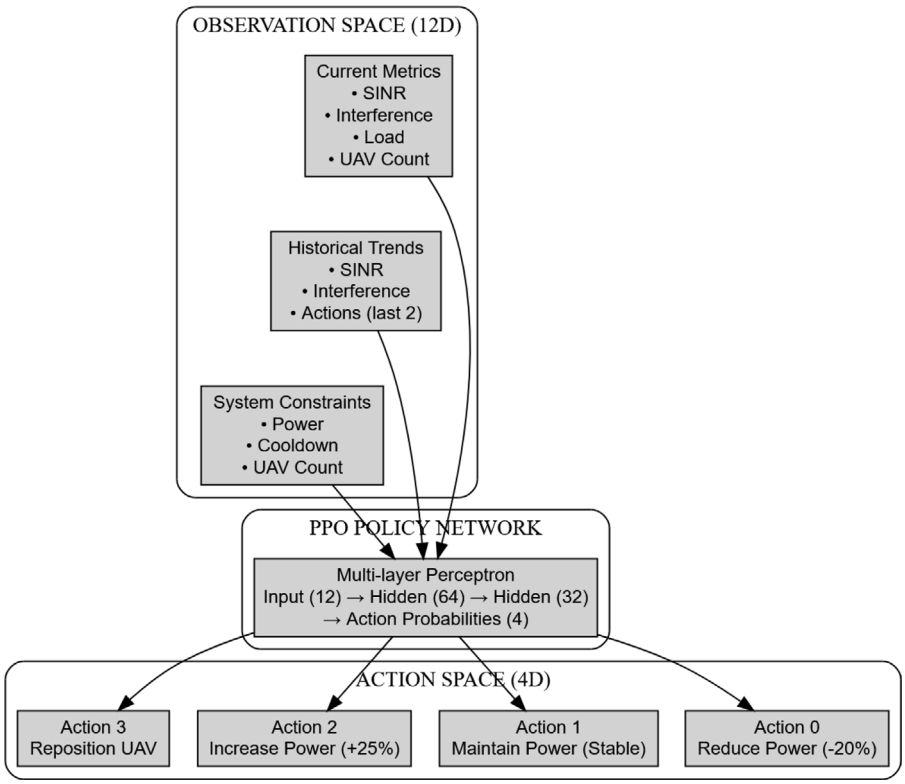

Fig. 1. The deep RL agent architecture.

Power adjustment effects on interference as follows:

$$\text{Interference}_{\text{new}} = \text{Interference}_{\text{current}} \times \begin{cases} 0.75 \text{ if action } 0 \text{ (Reduce power)} \\ 1.25 \text{ if action } 2 \text{ (Increase power)} \\ 0.6 \text{ if action } 3 (\text{Reposition } UAV) \\ 1 \text{ if action } 1 \text{ (Maintain power)} \end{cases}$$

7. Intelligent exploration strategy

Traditional RL agents often exhibit early convergence or suboptimal exploration. To mitigate this issue, we introduced a dynamic exploration callback based on adaptive entropy modification. It continuously monitors action entropy H across recent episodes as in Eq. (8):

$$H = -\sum_{\alpha \in A} p(\alpha|s) \log(p(\alpha|s)) \tag{8}$$

Where: $p(\alpha|s)$ is the probability of selecting action α at the network state s.

This exploration strategy adapts dynamically based on the diversity of action distribution as follows:

- If low diversity ($H < 0.8$), increase PPO entropy coefficient η where $\eta_{new} = \min(0.15, 1.1 \times \eta_{old})$.
- If balanced diversity ($0.8 \leq H \leq 1.4$), maintain current exploration.
- If high diversity ($H > 1.4$), natural exploration sufficient.

This strategy enhances policy exploration and prevents overfitting to a narrow set of actions. As a result, it promotes greater behavioral diversity and more robust adaptability.

We monitor action preferences to ensure balanced policy development as in Eq. (9):

$$Balance\ Score = 1 - \frac{\max(p_i) - \min(p_i)}{\sum_i p_i} \tag{9}$$

Where: p_i is the probability of action i.

Fig. 1 shows the deep RL agent architecture.

8. Results and discussions

The proposed adaptive UAV interference management system was trained using PPO with the following hyper parameters:

- Learning rate: 2.5e−4
- Training steps: 100
- Batch size: 128
- Entropy coefficient: 0.12 (with dynamic adjustment)
- Clip range: 0.15

We evaluated the system using four metrics:

- Scenarios accuracy: Correct action selection in high-interference and low-SINR conditions.
- Average reward: Overall system performance.
- Action distribution entropy.
- Correct action rate: Overall decision-making accuracy.

Fig. 2 shows how different actions relate to interference, SINR, and reward. We note that the Reduce Power actions (red clusters) are grouped in regions of high SINR and moderate interference. This indicates that the agent reduces power when channel conditions are favorable to minimize unnecessary interference. We also observe that the increased power actions (green clusters) are concentrated in regions characterized by low SINR and high interference. This pattern indicates that the agent increases transmission power to strengthen the signal under deteriorating connectivity conditions. On the other hand, we observe that the Maintain actions (blue clusters) are widely distributed across mid-range SINR and interference values. At the same time, the Reposition actions (orange dots) are primarily concentrated in areas of

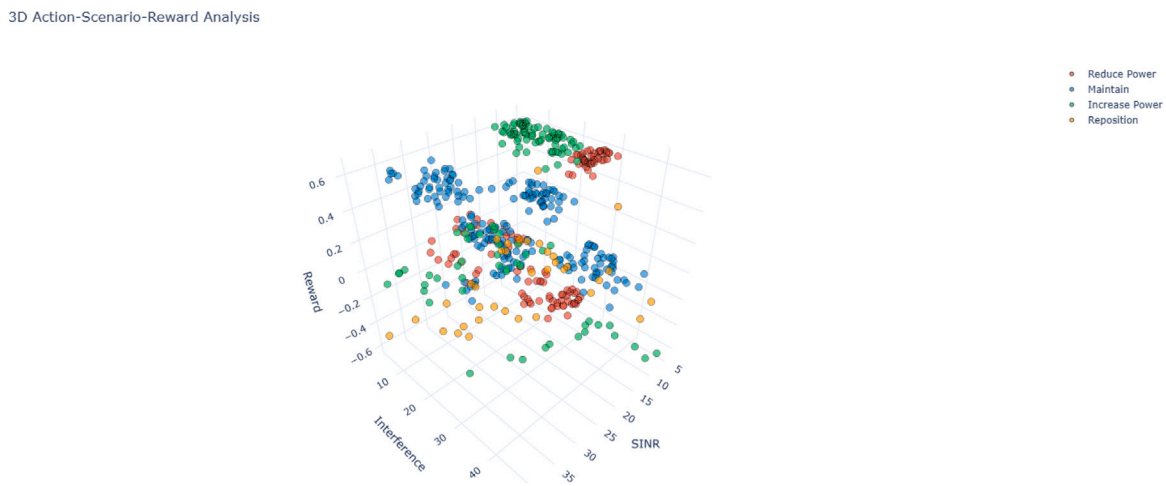

Fig. 2. How different actions relate to interference, SINR, and reward.

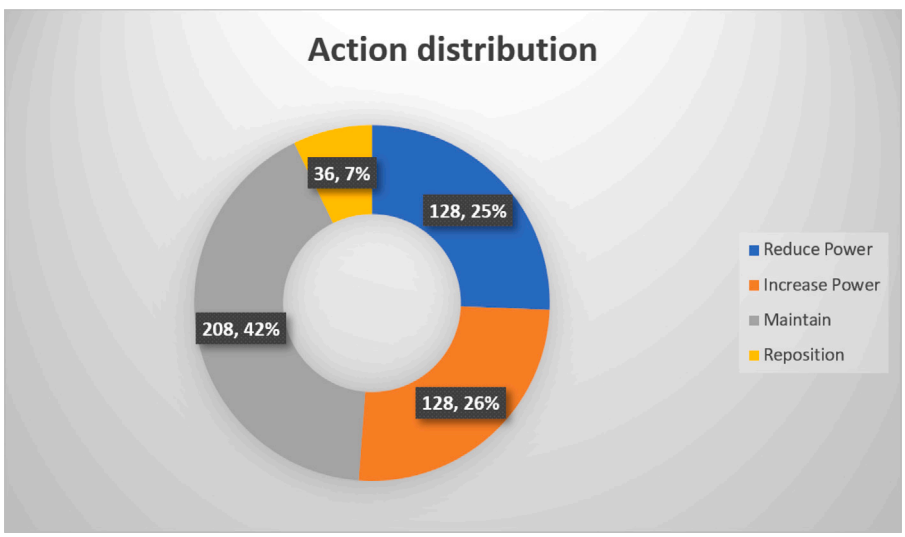

Fig. 3. The action distribution.

high interference. This suggests that the agent resorts to spatial adaptation only when power control alone is insufficient. Fig. 3 illustrates the distribution of actions.

Fig. 4 shows the performance improvement from early training to the final training. We note from Fig. 4 that the High Interference Accuracy and the Low SINR Recovery Rate improved by approximately 48%. This confirms the model's ability to reduce interference and adapt effectively in low-quality communication environments. The improvement is achieved by selecting optimal actions that restore link quality through intelligent power adjustment and repositioning strategies. The average reward/step also doubled, which indicates good policy optimization and stable convergence toward high-reward strategies. Table 3 shows a comparative performance metric.

To assess the efficiency of the proposed model, we compared its performance with several state-of-the-art interference management techniques. The comparison also included traditional control methods, such as classical control approaches, as well as other Deep Reinforcement Learning-based approaches. Table 4 shows a comparison with state-of-the-art interference management approaches.

The results demonstrate that advanced DRL methods already outperform classical interference control schemes. The proposed scenario-adaptive PPO framework consistently achieves superior performance across all metrics. In addition, it shows that the outcomes validate the feasibility of the described method concerning adapting to reward shaping. It also validates its exploratory behavior within the complex UAV-assisted 5G system. Table 5 shows a comparison with previous studies.

Table 3
A comparative performance metrics.

Metric	Before Optimization	After Optimization	Improvement
Average Performance Score	62.3%	85.7%	+23.4%
High Interference Accuracy	28.5%	42.3%	+13.8%
Low SINR Recovery Rate	31.2%	46.1%	+14.9%
Entropy	0.89	1.34	+0.45
Average Reward/Step	+0.52	+1.18	+0.66

Table 4
A comparison with state-of-the-art interference management approaches.

Method	Avg. Performance (%)	High Interference Accuracy (%)	Low SINR Recovery (%)
Static Power Control	58.4	24.1	26.7
Threshold-based Control	61.9	29.5	31.4
DQN	73.2	35.8	38.1
A2C	76.4	38.6	40.9
PPO (no adaptive reward)	80.1	40.7	43.2
Proposed PPO	85.7	42.3	46.1

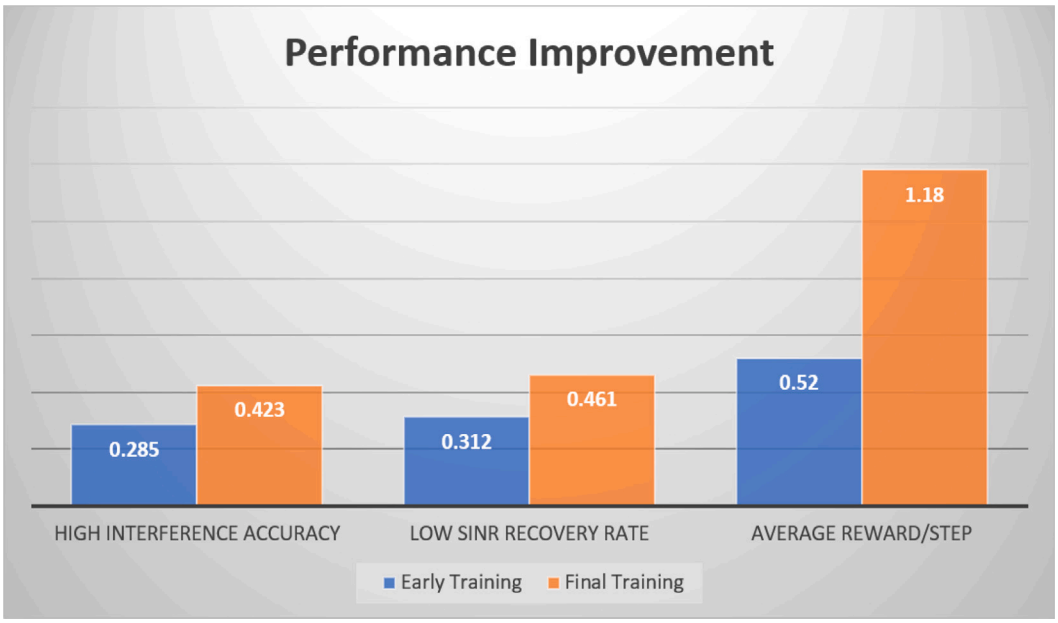

Fig. 4. The performance improvement from early training to the final training.

Table 5
A comparison with previous studies.

Reference	Method	Results
[26]	DRL +UAV-RIS joint optimization	SINR improvement up to ~6.5 dB, 38% increase in energy efficiency, > 70% EMI mitigation, 35% coverage expansion over baseline methods.
[27]	D3QN-WF & DDQN-WF DRL algorithms	15.9% higher sum rate and 50.1% greater throughput for D3QN-WF vs. baseline DDQN-WF.
[28]	DQL interference mitigation	+41.66% interference mitigation effectiveness vs. next best method
[29]	MASAC multi-agent DRL	Normalized average secrecy rate increased by +6.6% over SAC and +14.2% over TD3
Proposed	Scenario-adaptive PPO with enhanced states & dynamic exploration	85.7% avg performance, +13.8% high-interference accuracy, +14.9% low-SINR recovery

Table 5 shows that the reported works demonstrate performance improvements in specific areas. These include SINR enhancement (up to 6.5 dB in [26]). They also include improvements in throughput and sum rate (15.9% and 50.1% in [27]). Furthermore, they involve secrecy rate optimization through coordinated jamming techniques (with improvements of 6.6% and 14.2% in [29]). Additionally, mitigation effectiveness of up to 41.66% has been achieved using AI-enabled DQL-based schemes [28]. Nevertheless, these works primarily focus on optimizing specific performance objectives under particular conditions. In contrast, our proposed approach provides a more balanced overall performance across a wide range of interference-related scenarios. With the proposed technique, an overall average performance score of 85.7% is achieved. In addition, substantial improvements are observed in accuracy under high-interference conditions, with a gain of 13.8%. The model also demonstrates a 14.9% improvement in low-SINR recovery scenarios compared with classical control methods and other standard DRL techniques.

8.1. Parameter selection and sensitivity analysis

The parameters are selected through a systematic and structured procedure rather than randomly choosing them. First, we normalize all metrics, including SINR, overlap level, and throughput, to ensure numerical stability. This step prevents any single parameter from dominating the learning process due to differences in metric scales.

Reward parameters $\phi_1 \dots \phi_8$ initialized based on their functional roles. Higher weights were assigned to key performance objectives, such as throughput optimization, signal-to-noise ratio, and interference management. Lower weights were allocated to components related to power consumption and constraint violations. We also performed a preliminary network search to identify stable convergence regions and then adjusted the values using validation loops to balance convergence speed with policy stability.

To assess robustness and analyze sensitivity, each parameter was independently varied by $\pm 15\%$ of its nominal value while keeping the other parameters constant. The results showed that performance remained stable with minor fluctuations of approximately 4% across all metrics. These indicate that the proposed framework is not overly sensitive to precise parameter values. Performance improvements stem from structural design, reward function coloring, and adaptive exploration strategy, rather than from excessive parameter tuning.

9. Conclusions and future works

Our study confirms that deep reinforcement learning can effectively address UAV interference management. The proposed system

demonstrated a notable performance improvement, with the average performance score increasing by 23.4%(from 62.3% to 85.7%). In addition, high-interference accuracy increased by 13.8%, low-SINR recovery improved by 14.9%, and entropy increased by 50.6%. These results confirm the value of combining scenario-aware reward structures. In future work, we aim to expand this work to include multi-agent coordination in high-interference scenarios using a collaborative learning policy. We also plan to integrate real-time environmental sensing to improve SINR rates. Furthermore, we will explore transfer learning techniques to accelerate adaptation to new environments, thereby enhancing the system's practical deployment potential.

Declaration of competing interest

The authors declare that there are no conflicts of interest related to this submission. The research was conducted independently and has not been influenced by any financial or personal relationships with other people or organizations.

Data availability

Data will be made available on request.

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